Blue Mountains Bushfire Prediction

Machine Learning Report Contribution

Study design and inputs

We built a research prototype for next-day eligible satellite fire detection in Blue Mountains National Park. The frozen model and its two development-selected recall policies did not establish reliable warning performance. The deliverable is an auditable ML comparison, not a deployed warning system or a validated ignition-probability map.

Prediction unit and evidence

Each row is one retained ERA5-Land grid cell and one UTC feature day; the target is at least one eligible VIIRS detection on the following UTC day. Continuing fire counts as positive. Non-detection is not proof of no physical fire. Weather is aggregated over the grid cell, while terrain and labels concern its park portion; the group data owner must retain source and geometry documentation. ERA5-Land is retrospective reanalysis [6].

Temporal separation

Partition	UTC feature dates	Rows
Training	1 July to 8 December 2019	6923
Gap	9 to 15 December 2019	301
Validation	16 to 24 December 2019	387
Gap	25 to 31 December 2019	301
Final holdout	1 January to 29 June 2020	7,783

All cells from a date stayed together. The supplier's future-test boundary was preserved, with seven-day gaps rather than interleaved weeks [3]. At least 10% park overlap retained 43 cells. Rainfall warm-up excluded the first two feature days from fitting, leaving 6,837 training rows and 143 positive targets. Gaps reduce adjacency, not dependence between long-running fires.

Reviewed inputs

The core has sixteen weather/terrain variables; the second version adds only the audited same-day fire_today flag. Target-day counts, labels, dates, grid identifiers and coordinates are excluded as predictors. Long rainfall/dry-history and other current-fire candidates were not enabled because their warm-up or semantics needed further treatment. Fuel-moisture imagery and NDVI were not implemented.

Models and evaluation procedure

Matched models and stopping

Training prevalence and persistence were explicit references. Logistic regression and XGBoost were each compared with sixteen and seventeen inputs on matched fitting/evaluation rows. Logistic scaling and balanced weights used training rows only. Tree fitting used training negatives divided by positives for class emphasis; calibration did not rebalance labels. XGBoost tests nonlinear thresholds/interactions [1], rather than guaranteeing better performance.

XGBoost setting	Recorded value
Maximum depth	3
Learning rate	0.05
L2 leaf penalty	5.0
Row and feature subsampling	0.8 and 0.9
Early stopping	25 rounds without AP improvement
Final 16 and 17 input tree counts	6 and 8

Average precision (AP) was computed from raw scores using sklearn, and explicitly maximised for stopping [2]. The initial validation selected tree length and was development evidence. Later forward checks selected length on past inner data and refitted that fixed count before checking later dates. Missing-class windows remained visible; the dependent windows were not treated as independent replication.

Probability and reminder decisions

A nondecreasing regularised sigmoid was fitted on base-model-excluded calibration rows, then checked on later dates, with seven-day gaps and no post-calibration base refit. The week-long design worsened later probability quality; the improving recent design had only one calibration and one assessment day. Calibration was therefore not adopted [4]. The highest calibration-only cutoff meeting each requested recall constraint was retained, with ties together.

Development target	Exact raw cutoff	Comparison
80.0%	0.6658568382263184	>=
90.0%	0.4193871319293976	>=

The complete source, features, artifacts, thresholds and reporting rules were committed in freeze ef45cdf before one final release [7]. No final fitting, calibration or threshold search occurred. AP was primary; confusion counts, probability quality and alert burden were reported overall, without current detections, and for every calendar month.

Final findings

The final period has 7,783 cell-days, 181 dates and 49 positive next-day targets (0.63%); these are not independent fires. Both frozen development recall targets failed to transfer [7].

Frozen policy	TP	FP	FN	Recall	Precision
80% target	16	39	33	32.7%	29.1%
90% target	34	2,434	15	69.4%	1.4%
Persistence	16	39	33	32.7%	29.1%

Model at diagnostic 0.5	AP	Recall	Brier
Training prevalence	0.0063	0.0%	0.0065
Persistence	0.0992	32.7%	0.0093
Logistic 16	0.0461	81.6%	0.1319
Logistic 17	0.2113	63.3%	0.0391
XGBoost 16	0.0219	57.1%	0.1912
XGBoost 17	0.1062	32.7%	0.1511

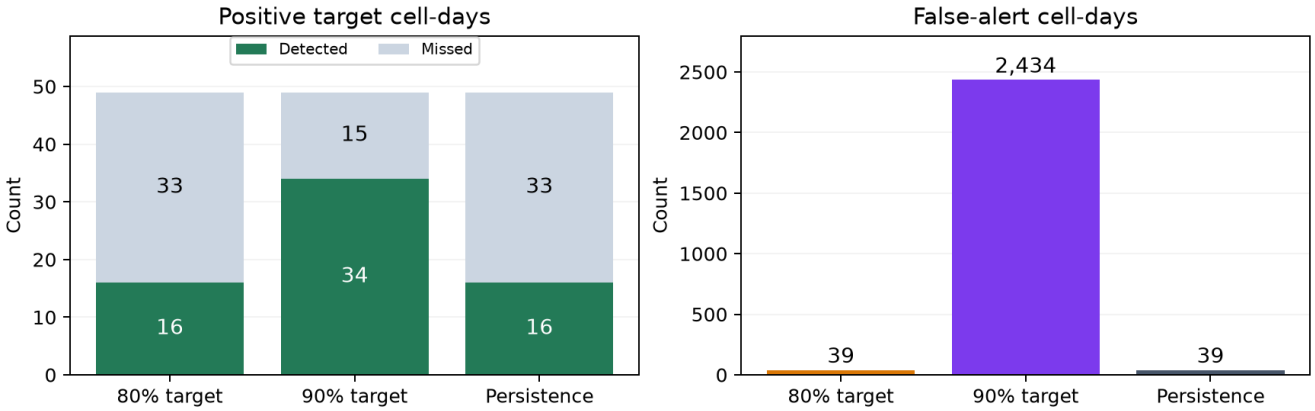


Figure 1 Frozen-policy trade-off. Counts are cell-days, not incidents or staffing requirements. The development target is not a promise of final recall.

The prespecified logistic 17 comparison ranked better than XGBoost 17; this did not trigger a replacement or retuning. Without current detections, the higher cutoff found 0 of 33 positives; the lower found 18 with 2,395 false alerts. All months were retained, including zero-positive months with undefined AP/recall. A no-alert reference scored 99.37% accuracy while missing every positive.

Reliability and implications

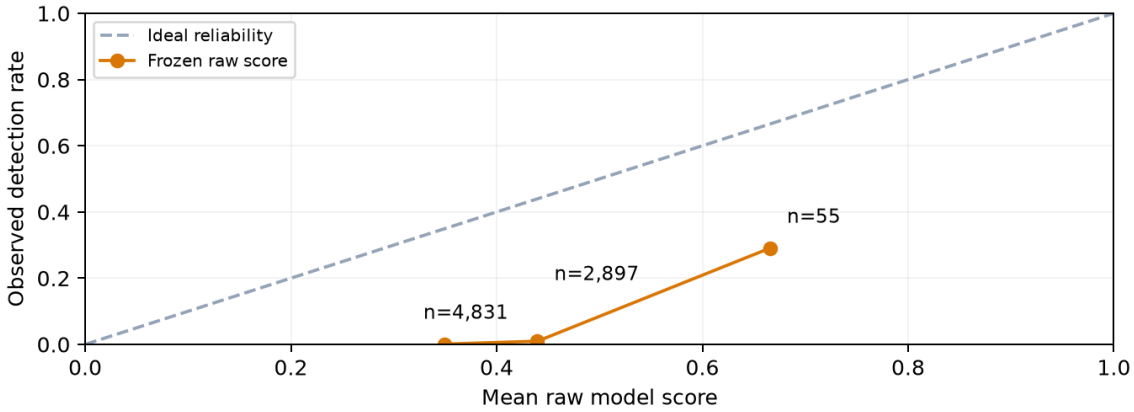


Figure 2 Fixed-bin reliability for the frozen raw score. Bin sizes and observed frequencies do not establish independent-event precision or calibrated percentages.

The raw scores overstate observed next-day detection frequencies. The small AP advantage over persistence does not establish practical warning benefit, and Brier/log loss measure overall probability quality rather than calibration alone. We did not fit a correction using final outcomes [4,7].

Interpretation and recommendations

The evidence supports a research prototype, not operational dispatch or public risk percentages. False alerts could waste review capacity and erode trust; missed detections could create false reassurance. A future interface should retain human review, label score meaning, provenance and timing, and distinguish satellite non-detection from physical safety. Response capacity and the costs of false versus missed alerts require stakeholder agreement, not invented financial assumptions.

Transfer is limited by sparse early positives, changing prevalence, one supplied season, dependent cells/events, imperfect detection coverage and coarse spatial aggregation. Retrospective publication latency was not simulated. The related Australian multi-source study motivates environmental modelling, but its task, data and headline accuracy are not a like-for-like benchmark for our next-day AP evaluation [5].

Further work needs independent seasons/events, checked observation coverage, suitable fuel information and forecast-time data availability before calibration and response-policy validation. This exposed test must not become a tuning set. The ML contribution completes the auditable comparison; the group must still verify source/licence/map details, domain interpretation, citations, authorship and AI-use records before submission.

References

[1] Tianqi Chen and Carlos Guestrin. XGBoost A Scalable Tree Boosting System. 2016. Source.

[2] scikit-learn developers. Average precision score. 2026. Source.

[3] scikit-learn developers. Cross validation of time series data. 2026. Source.

[4] scikit-learn developers. Probability calibration. 2026. Source.

[5] Tanvi Jois and Hussain Ahmad and Fatima Noor and Faheem Ullah. Australian Bushfire Intelligence with AI Driven Environmental Analytics. 2026. Source.

[6] Copernicus Climate Change Service. ERA5 Land hourly data from 1950 to present. 2026. Source.

[7] ENGG2112 project team. Blue Mountains frozen final temporal evaluation. 2026. Source.